Timeline Check-In

Set Up

Packages

```r
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# read .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)

library(dplyr)     # data manipulation
library(broom)     # tidy statistical output
library(corrplot)
library(ggrepel)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")


# read in site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# read in water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

Cleaning

`sites` object

```
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

Cleaning `taxon_list`

```
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

Cleaning `water_quality`

```r
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(dissolved_oxygen_mg_l, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

Cleaning `aq_ins`

```r
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
```

```r
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (sum abundance divided by 7.5 liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(water_quality_clean, by = c("date_site")) |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_do, .fun = "median", .na_rm = TRUE),
         site_full = fct_rev(site_full))
```

Focus Taxa (`focus_taxa`)

```r
# taxa to include in all figures
focus_taxa <- c("Ostracod", "Copepod", "Cladocera",
                "Hemiptera Corixidae Boatman", "Nematode")
```

Creating DO bins (creating `do_category` column)

```
aq_ins_clean_bins <- aq_ins_clean |>
  mutate(bins = case_when(site_full == "Phelps Road" ~ "low",
                          site_full == "North of Phelps Bridge" ~ "low",
                          site_full == "South of Venoco Bridge" ~ "moderate",
                          site_full == "South of Phelps Bridge" ~ "moderate",
                          site_full =="Devereux Creek" ~ "moderate",
                          site_full =="North of Venoco Bridge" ~ "high",
                          site_full == "Main Channel" ~ "high")) |>
  mutate(taxon = str_replace_all(taxon, "_", " "), taxon = str_to_title(taxon)) |>
  filter(taxon %in% focus_taxa) |>
  # create column that calculates log-transformed abundance
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  mutate(bins = factor(bins, levels = c("low", "moderate", "high")))

# create object DO_taxa from aq_ins_clean
DO_taxa <- aq_ins_clean |>
  # remove rows that are missing median salinity values
  filter(!is.na(med_do)) |>
  # create column that calculates log-transformed abundance
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # mutates to title-case taxon names
  mutate(taxon = str_replace_all(taxon, "_", " "), taxon = str_to_title(taxon)) |>
  # only include focus taxa
  filter(taxon %in% focus_taxa) |>
  # order taxon levels by maximum average abundance per liter
  mutate(taxon = fct_reorder(taxon, ave_lit, .fun = max, .desc = TRUE)) |>
  # keep only the columns needed for the figure
  select(date, site, site_full, taxon, ave_lit, log_ave_lit, med_do)
total_abundance <- DO_taxa |>
  filter(taxon %in% focus_taxa) |>
  group_by(date, site, med_do) |>
  summarize(
    total_abundance = sum(ave_lit, na.rm = TRUE),
    log_total_abundance = log10(total_abundance + 1),
    .groups = "drop"
  )
total_abundance <- total_abundance |>
  mutate(
    do_category = case_when(
      med_do <= quantile(med_do, 0.25, na.rm = TRUE) ~ "Low DO",
```

```
      med_do <= quantile(med_do, 0.75, na.rm = TRUE) ~ "Medium DO",
      TRUE ~ "High DO"
    )
  )
total_abundance$do_category <- factor(
  total_abundance$do_category,
  levels = c("Low DO", "Medium DO", "High DO")
)
```

Ideas for analysis and background

Ideas for analysis:

Our first big idea is looking at total invertebrate abundance compared to median dissolved oxygen levels. We ran a correlation test and found that there was a statistically significant negative linear relationship between median dissolved oxygen and log-transformed total invertebrate abundance, suggesting that abundance decreased as DO increased.

We also explored how dissolved oxygen levels differ between sites, and then categorized sites into low, moderate, and high DO bins. We found that total log-transformed invertebrate abundance did not differ significantly across low, moderate, and high DO bins. However, Ostracod abundance differed significantly across DO bins, indicating that Ostracods may respond differently to low, moderate, and high DO conditions. After Bonferroni correction, the moderate vs. high DO comparison was close to significant, suggesting a possible difference in Ostracod abundance between these bins but not strong enough to conclude significance at $= 0.05$.

Another way we compared taxa in bins was by running a Chi-square test on taxon composition within each DO bin. We found that taxon composition differed significantly across DO bins, suggesting that the invertebrate community structure changes depending on dissolved oxygen category. Next, we'd like to perform post-hoc tests to see which groups differ from one another.

Background:

North Campus Open Space is a restored wetland, which makes it a great area to explore data about the effects of restoration. One metric of restoration progress is water quality, and dissolved oxygen concentration is a key water-quality indicator. We are interested in the relationship between invertebrate abundance and dissolved oxygen, which ultimately ties back to the health of this wetland. This is a less traditional method of measuring ecosystem health, but can become more widely used with increasing data showing these relationships. This study applies to all wetland ecosystems and aquatic environments more broadly. In addition to being a metric of ecosystem health, invertebrate abundance compared to dissolved oxygen can tell us about the conditions that these invertebrates exist in.

Citations:

Poster: Algae, Macroinvertebrate and Water Quality Relationships at North Campus Open Space - link

Background Articles

Connolly, N. M., Crossland, M. R., & Pearson, R. G. (2004). *Effect of low dissolved oxygen on survival, emergence, and drift of tropical stream macroinvertebrates.* BioOne Complete (BioOne). https://doi.org/10.1899/0887-3593(2004)023%3C0251:eoldoo%3E2.0.co;2

Gunsch, D. E. (2020, November 5). *Community Composition of Aquatic Invertebrates along Dissolved Oxygen Gradients in Lake Huron Coastal Wetland Wet Meadows.* Handle.net. https://hdl.handle.net/20.500.14776/9073

Lucchesi, D. O., Chipps, S. R., & Schumann, D. A. (2022). *Effects of seasonal hypoxia on macroinvertebrate communities in a small reservoir.* Lakes & Reservoirs: Science, Policy and Management for Sustainable Use, 27(1). https://doi.org/10.1111/lre.12395

Suren, A. M., Lambert, P., & Sorrell, B. K. (2010). *The Impact of Hydrological Restoration on Benthic Aquatic Invertebrate Communities in a New Zealand Wetland.* Restoration Ecology, 19(6), 747–757. https://doi.org/10.1111/j.1526-100x.2010.00723.x

Spearman Correlation Analysis for Total Abundance and DO levels

```
cor.test(
  total_abundance$log_total_abundance,
  total_abundance$med_do,
  method = "spearman",
  exact = FALSE)
```

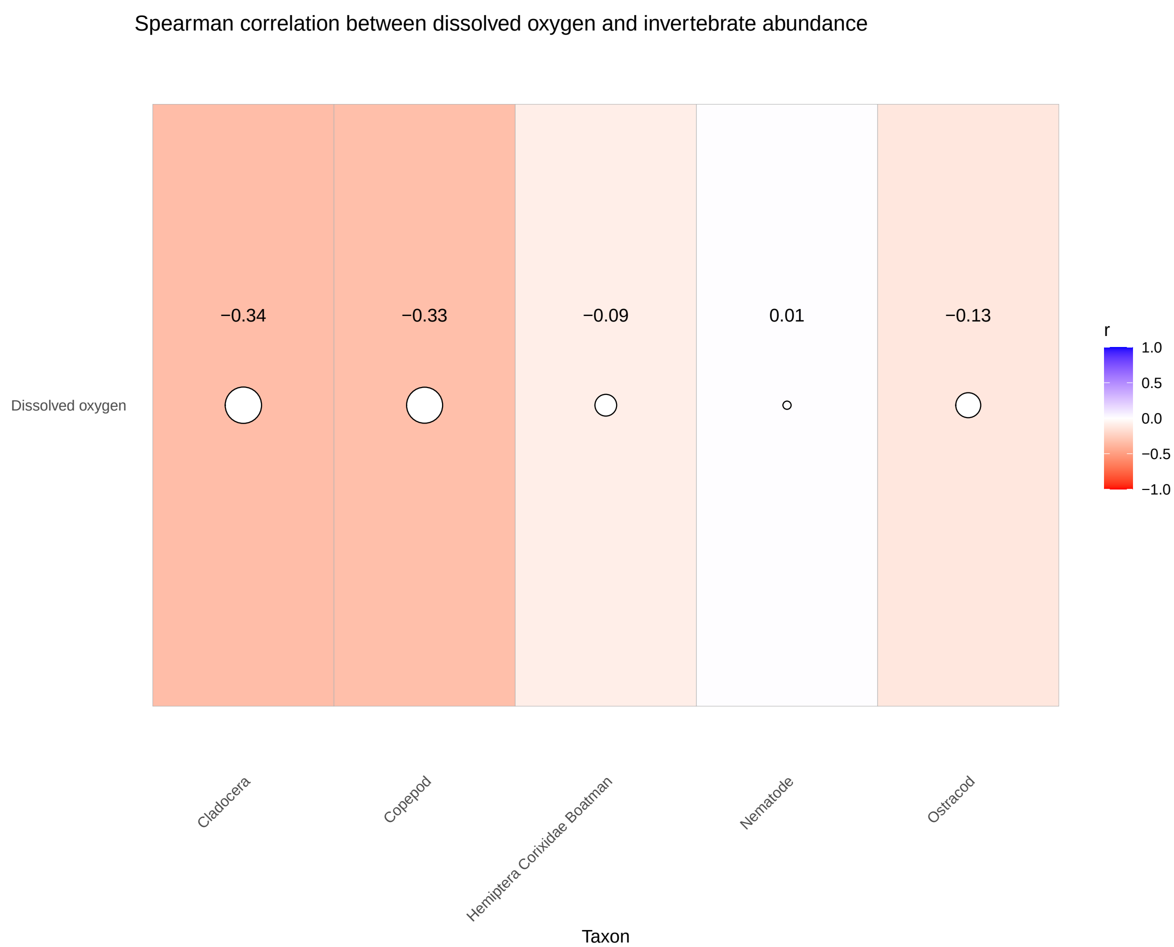
```

	Spearman's rank correlation rho

data:  total_abundance$log_total_abundance and total_abundance$med_do
S = 8526.3, p-value = 0.08182
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
-0.3027201
```

Total log-transformed invertebrate abundance had a **weak negative** relationship with dissolved oxygen, but this relationship was not statistically significant using Spearman correlation.

Correlation analysis for Each Focus Taxa and DO levels

```
# create wide dataset (one column per taxon)
taxa_wide <- DO_taxa |>
  select(date, site, taxon, log_ave_lit, med_do) |>
  pivot_wider(
    names_from = taxon,
    values_from = log_ave_lit
  )

# calculate correlation of each taxon with dissolved oxygen
# calculate Spearman correlation of each taxon with dissolved oxygen
cor_do <- taxa_wide |>
  select(-date, -site) |>
  summarize(across(-med_do, ~ cor(.x,
                                  med_do,
                                  method = "spearman",
                                  use = "complete.obs"))) |>
  pivot_longer(cols = everything(),
               names_to = "taxon",
               values_to = "correlation")

ggplot(cor_do, aes(x = taxon, y = "Dissolved oxygen", fill = correlation)) +
  geom_tile(color = "grey70") +
  geom_point(aes(size = abs(correlation)), shape = 21, fill = "white") +

  # move labels ABOVE circles
  geom_text(aes(label = round(correlation, 2)),
            nudge_y = 0.15,  # <-- this moves text up
            size = 4) +

  scale_fill_gradient2(
    low = "red",
    mid = "white",
    high = "blue",
    midpoint = 0,
    limits = c(-1, 1),
```

```r
    name = "r"
  ) +
  scale_size(range = c(2, 10), guide = "none") +
  labs(
    x = "Taxon",
    y = NULL,
    title = "Spearman correlation between dissolved oxygen and invertebrate abundance"
  ) +
  theme_minimal() +
  theme(
    panel.grid = element_blank(),
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
```



Most focus taxa showed **weak negative** correlations with dissolved oxygen, with Cladocera and Copepod showing the strongest negative relationships and Nematode showing almost no relationship.

Pearson correlation for continuous DO and total community abundance

```
# Pearson correlation for continuous DO and total community abundance
cor.test(total_abundance$log_total_abundance,
         total_abundance$med_do,
         method = "pearson")
```

```
    Pearson's product-moment correlation

data:  total_abundance$log_total_abundance and total_abundance$med_do
t = -2.2882, df = 32, p-value = 0.02887
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.63289820 -0.04217141
sample estimates:
       cor
-0.3749894
```

```
# r = -0.3749894, p = 0.02887
# there is a significant negative correlation between abundance and continous DO
```

There was a **statistically significant negative linear relationship** between median dissolved oxygen and log-transformed total invertebrate abundance, suggesting that abundance decreased as DO increased.

Kruskal-Wallis test comparing abundance across DO categories

```
kruskal.test(log_total_abundance ~ do_category,
             data = total_abundance)
```

```
    Kruskal-Wallis rank sum test

data:  log_total_abundance by do_category
Kruskal-Wallis chi-squared = 2.0001, df = 2, p-value = 0.3679
```

```
# Kruskal-Wallis chi-squared = 3.2212, df = 2, p-value = 0.1998
# there is not a significant difference between abundance and DO categories
```

Total log-transformed invertebrate abundance **did not differ significantly** across low, moderate, and high DO categories.

Kruskal-Wallis test between abundance and bins for Copepods

```
kruskal.test(log_ave_lit ~ bins,
             data = aq_ins_clean_bins,
             subset = taxon == "Copepod")
```

```

	Kruskal-Wallis rank sum test

data:  log_ave_lit by bins
Kruskal-Wallis chi-squared = 4.405, df = 2, p-value = 0.1105
```

```
# Kruskal-Wallis chi-squared = 4.405, df = 2, p-value = 0.1105
# no significant difference
```

Copepod abundance **did not differ significantly across DO bins**, suggesting DO category was not a strong predictor of Copepod abundance.

Kruskal-Wallis test between abundance and bins for Ostracods

```
kruskal.test(log_ave_lit ~ bins,
             data = aq_ins_clean_bins,
             subset = taxon == "Ostracod")
```

```

	Kruskal-Wallis rank sum test

data:  log_ave_lit by bins
Kruskal-Wallis chi-squared = 6.7235, df = 2, p-value = 0.03467
```

```
# Kruskal-Wallis chi-squared = 6.7235, df = 2, p-value = 0.03467
# significant difference of abundance btwn DO level bins
```

Ostracod abundance **differed significantly across DO bins**, indicating that Ostracods may respond differently to low, moderate, and high DO conditions.

Pairwise test between abundance and individual bins for Ostracods

```
bins_ostracods <- aq_ins_clean_bins |>
  filter(taxon == "Ostracod")
pairwise.wilcox.test(
  bins_ostracods$log_ave_lit,
  bins_ostracods$bins,
  p.adjust.method = "bonferroni")
```

```

    Pairwise comparisons using Wilcoxon rank sum test with continuity correction

data:  bins_ostracods$log_ave_lit and bins_ostracods$bins

         low   moderate
moderate 0.387 -
high     1.000 0.051

P value adjustment method: bonferroni
```

```
# p = 0.051 for comparison btwn moderate and high DO level bins
```

After Bonferroni correction, the moderate vs. high DO comparison was close to significant, suggesting a possible difference in Ostracod abundance between these bins but not strong enough to conclude significance at $\alpha = 0.05$.

Chi-Square

```r
# create abundance table for chi-square test
chi_square_table <- aq_ins_clean_bins |>
  filter(taxon %in% focus_taxa) |>
  group_by(bins, taxon) |>
  summarise(
    total_abundance = sum(ave_lit, na.rm = TRUE),
    .groups = "drop"
  ) |>
  pivot_wider(
    names_from = taxon,
    values_from = total_abundance,
    values_fill = 0
  ) |>
  column_to_rownames("bins") |>
  as.matrix()

view(chi_square_table)

# show table used in chi-square test
chi_square_table
```

```
         Cladocera   Copepod Hemiptera Corixidae Boatman  Nematode    Ostracod
low       980.2667  115.8667                   0.2666667 0.1333333 1202.933333
moderate  510.6667 1170.4000                  83.8666667 0.2666667 1057.866667
high      264.2667 1453.4667                   8.8000000 0.0000000    7.866667
```

```r
# chi-square test for differences in taxon composition across DO bins
chi_square_test <- chisq.test(chi_square_table)

# show chi-square result
chi_square_test
```

```

    Pearson's Chi-squared test

data:  chi_square_table
X-squared = 2859.2, df = 8, p-value < 2.2e-16
```

Taxon composition **differed significantly across DO bins**, suggesting that the invertebrate community structure changes depending on dissolved oxygen category.

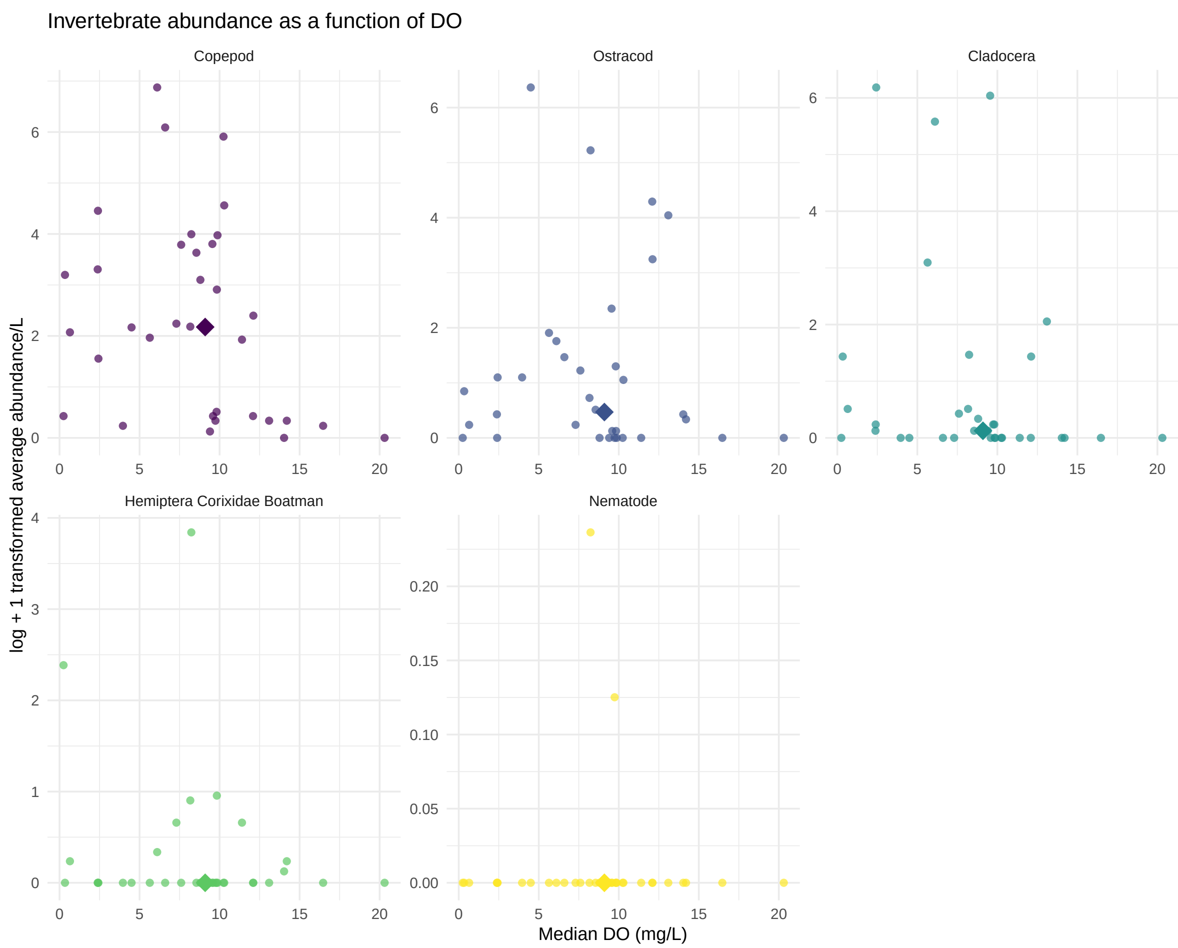
Visualizations directly relevant to answering questions

Directly Relevant Figure 1: DO levels vs invert. abundance for each focus taxa

```
DO_taxa_medians <- DO_taxa |>
  # groups data by taxon so each species gets one median point
  group_by(taxon) |>
  # calculates median DO and median log abundance for each taxon
  summarise(
    med_do_point = median(med_do, na.rm = TRUE),
    med_log_ave_lit = median(log_ave_lit, na.rm = TRUE),
    .groups = "drop"
)
```

```
DO_taxa_plot <- ggplot(data = DO_taxa, # assigns plot to object do_taxa_plot
                        # uses DO_taxa data frame
                        mapping = aes(x = med_do, # x-axis: med_do
                                      y = log_ave_lit, # y-axis: log_ave_lit
                                      color = taxon)) + # color by taxon
  # first layer: points show individual abundance measurements
  geom_point(shape = 16, # shape is circle
             size = 2, # size is 2
             alpha = 0.7) + # opacity is 0.7

  # second layer: larger point shows the median location for each taxon
  geom_point(data = DO_taxa_medians,
             mapping = aes(x = med_do_point,
                           y = med_log_ave_lit),
             shape = 18,
             size = 5) +

  # separate panels for each taxon with free x- and y-axis scales
  facet_wrap(~ taxon, scales = "free") +
  # use non-default colors for points
  scale_color_viridis_d() +
  labs(x = "Median DO (mg/L)", # relabel x-axis
       y = "log + 1 transformed average abundance/L", # relabel y-axis
       # creates descriptive title
       title = "Invertebrate abundance as a function of DO") +

  # uses non-default theme
```

```
  theme_minimal() +
  # remove redundant legend
  theme(legend.position = "none")
DO_taxa_plot
```



Description of visual components

This figure shows the relationship between Median DO (mg/L) and log + 1 transformed average invertebrate abundance/L for our five focus taxa. Each panel represents a different taxon. Each small point represents one observation of that taxon, with median dissolved oxygen shown on the x-axis in mg/L and log + 1 transformed average abundance per liter shown on the y-axis. The different colors distinguish the taxa, and the larger diamond in each panel represents the overall average/summary point for that taxon across observations.

Main message / takeaway

Overall, there does not appear to be a strong or consistent relationship between median dissolved oxygen and invertebrate abundance across all taxa. Some taxa, especially Copepods, Ostracods, and Cladocerans, show higher abundances at low to moderate dissolved oxygen levels, but the pattern is variable and differs by taxon.

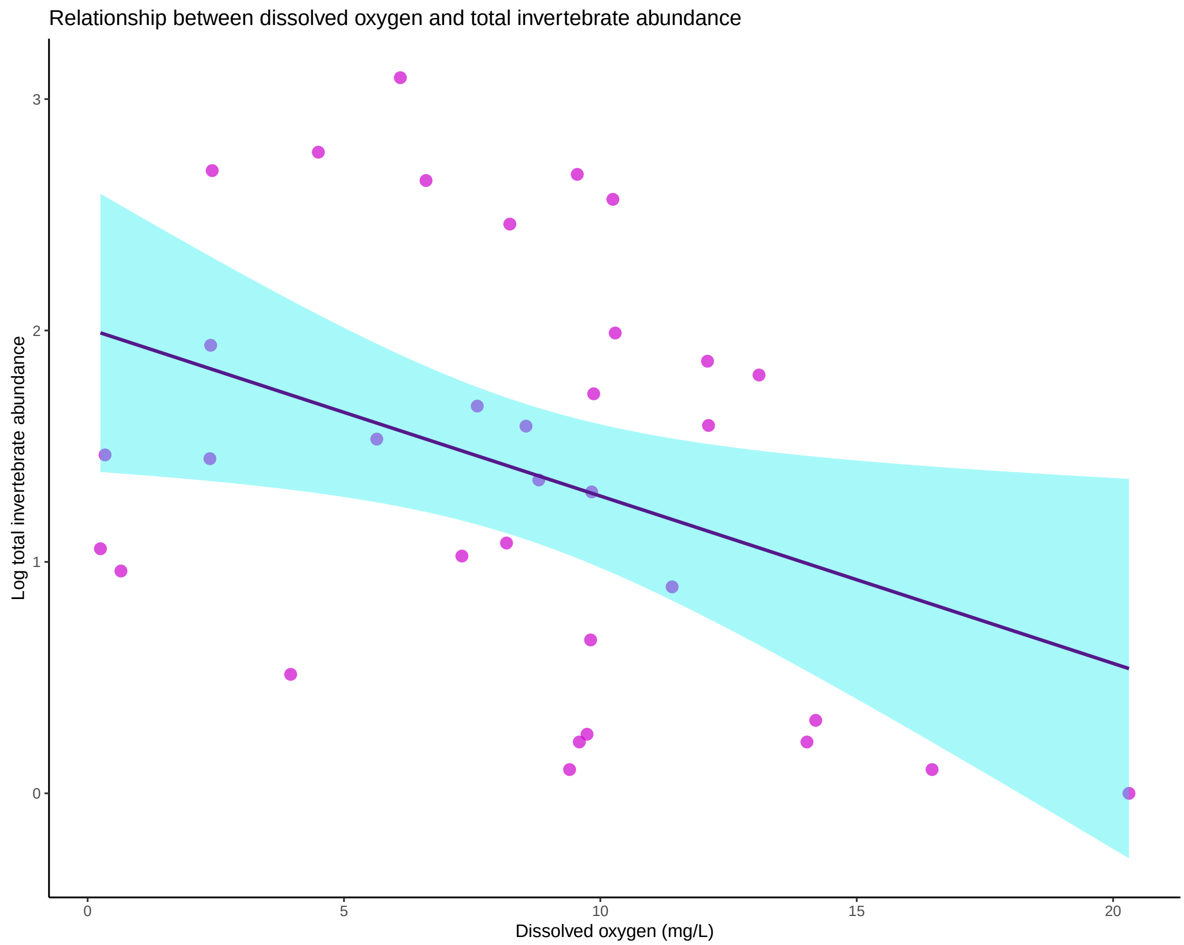
Connection to the main question

This figure helps evaluate whether dissolved oxygen is related to invertebrate abundance in the study system. The variation among taxa suggests that dissolved oxygen may influence some invertebrate groups differently, but DO alone does not clearly explain abundance patterns across the full invertebrate community.

Directly Relevant Figure 2: DO levels vs total invert. abundance

```
# Summed abundance across focus taxa vs DO

total_abundance <- DO_taxa |>
  filter(taxon %in% focus_taxa) |>
  group_by(date, site, med_do) |>
  summarize(
    total_abundance = sum(ave_lit, na.rm = TRUE),
    .groups = "drop"
  ) |>
  mutate(log_total_abundance = log10(total_abundance + 1))

# fit linear model
total_abundance_lm <- lm(log_total_abundance ~ med_do,
                         data = total_abundance)

# create new data frame for predicted values
total_abundance_predictions <- data.frame(
  med_do = seq(min(total_abundance$med_do, na.rm = TRUE),
               max(total_abundance$med_do, na.rm = TRUE),
               length.out = 100)
)

# generate predicted values and confidence intervals
total_abundance_predictions <- total_abundance_predictions |>
  bind_cols(
    predict(total_abundance_lm,
            newdata = total_abundance_predictions,
            interval = "confidence") |>
```

```
    as.data.frame()
  )

# plot raw data and model predictions
ggplot(total_abundance,
       aes(x = med_do,
           y = log_total_abundance)) +

  # raw data points
  geom_point(color = "magenta3",
             size = 3,
             alpha = 0.7) +

  # confidence interval around model predictions
  geom_ribbon(data = total_abundance_predictions,
              aes(x = med_do,
                  ymin = lwr,
                  ymax = upr),
              inherit.aes = FALSE,
              fill = "cyan2",
              alpha = 0.35) +

  # predicted line from linear model
  geom_line(data = total_abundance_predictions,
            aes(x = med_do,
                y = fit),
            inherit.aes = FALSE,
            color = "purple4",
            linewidth = 1) +

  labs(
    x = "Dissolved oxygen (mg/L)",
    y = "Log total invertebrate abundance",
    title = "Relationship between dissolved oxygen and total invertebrate abundance"
  ) +

  theme_classic()
```




```r
broom::tidy(total_abundance_lm)
```

```
# A tibble: 2 x 5
  term        estimate std.error statistic     p.value
  <chr>          <dbl>     <dbl>     <dbl>       <dbl>
1 (Intercept)   2.01      0.302       6.65 0.000000170
2 med_do       -0.0723    0.0316     -2.29 0.0289
```

Description of visual components

This figure shows the relationship between DO and total invertebrate abundance summed across our five focus taxa. Each pink point represents one sampling observation grouped by date, site, and median dissolved oxygen, where abundance was summed across the focus taxa and then log-transformed using log10(total_abundance + 1). The x-axis shows median dissolved oxygen in mg/L, and the y-axis shows log-transformed total invertebrate abundance.

The purple line represents the predicted relationship from a linear model, and the light blue shaded area represents the confidence interval around the model prediction.

Main message / takeaway

The figure shows a negative relationship between DO and total invertebrate abundance, where total invertebrate abundance tends to decrease as dissolved oxygen increases. However, there is still noticeable variation around the trend line, so dissolved oxygen explains some, but not all, of the variation in total abundance.

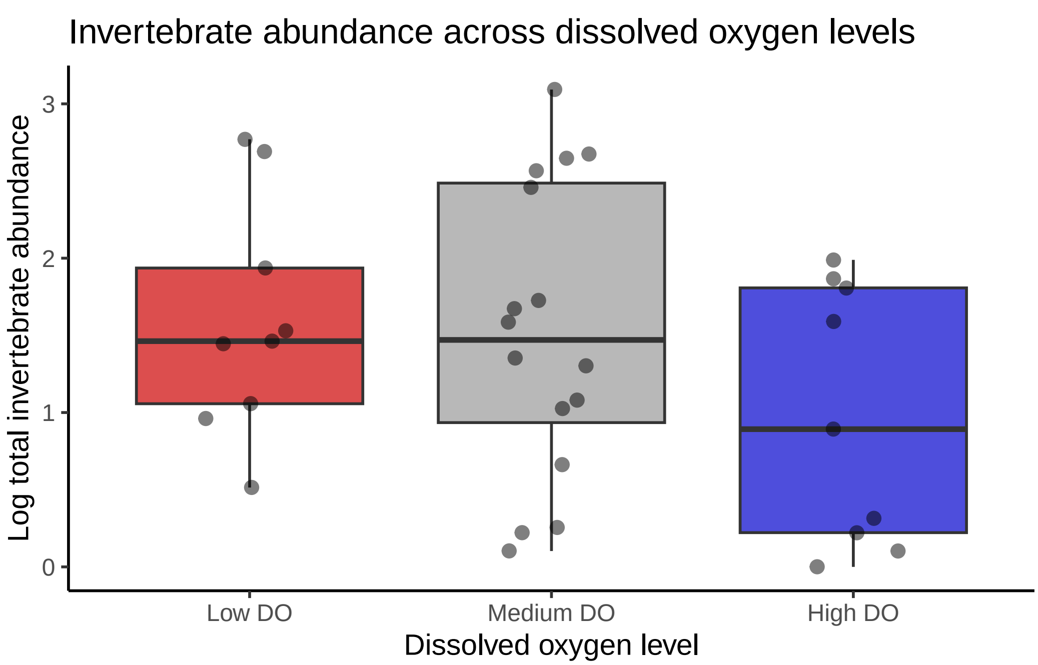
Connection to the main question

This figure directly addresses whether dissolved oxygen is related to invertebrate abundance by combining the focus taxa into one community-level abundance measure. It helps contextualize the main question by showing that lower dissolved oxygen conditions are associated with higher invertebrate abundance in this dataset.

Directly Relevant Figure 3: Abundance vs DO with binned levels

```r
total_abundance <- total_abundance |>
  mutate(
    do_category = case_when(
      med_do <= quantile(med_do, 0.25, na.rm = TRUE) ~ "Low DO",
      med_do <= quantile(med_do, 0.75, na.rm = TRUE) ~ "Medium DO",
      TRUE ~ "High DO"
    )
  )
total_abundance$do_category <- factor(
  total_abundance$do_category,
  levels = c("Low DO", "Medium DO", "High DO")
)

ggplot(total_abundance,
       aes(x = do_category,
           y = log_total_abundance,
           fill = do_category)) +

  geom_boxplot(alpha = 0.7,
               outlier.color = "black",
               outlier.size = 2) +

  # optional: add jittered points to show raw data
  geom_jitter(width = 0.15,
```

```
             alpha = 0.5,
             size = 2,
             color = "black") +

labs(
  x = "Dissolved oxygen level",
  y = "Log total invertebrate abundance",
  title = "Invertebrate abundance across dissolved oxygen levels"
) +

scale_fill_manual(values = c(
  "Low DO" = "red3",
  "Medium DO" = "grey60",
  "High DO" = "blue3"
)) +

theme_classic() +
theme(legend.position = "none")
```



Description of visual components

This figure compares log-transformed total invertebrate abundance across three DO categories: Low DO, Medium DO, and High DO. Each black jittered point represents one sampling observation grouped by date, site, and median dissolved oxygen, where abundance was summed across the five focus taxa and transformed using log10(total_abundance + 1). The colored boxes show the middle 50% of abundance values for each DO category, the thick horizontal line inside each box represents the median, and the whiskers show the spread of values outside the interquartile range.

Main message / takeaway

Total invertebrate abundance appears lower in the High DO category compared with the Low DO and Medium DO categories. However, there is substantial overlap and variability among the DO categories, especially in the Medium DO group, so the difference is not extremely clear-cut from the visualization alone.

Connection to the main question

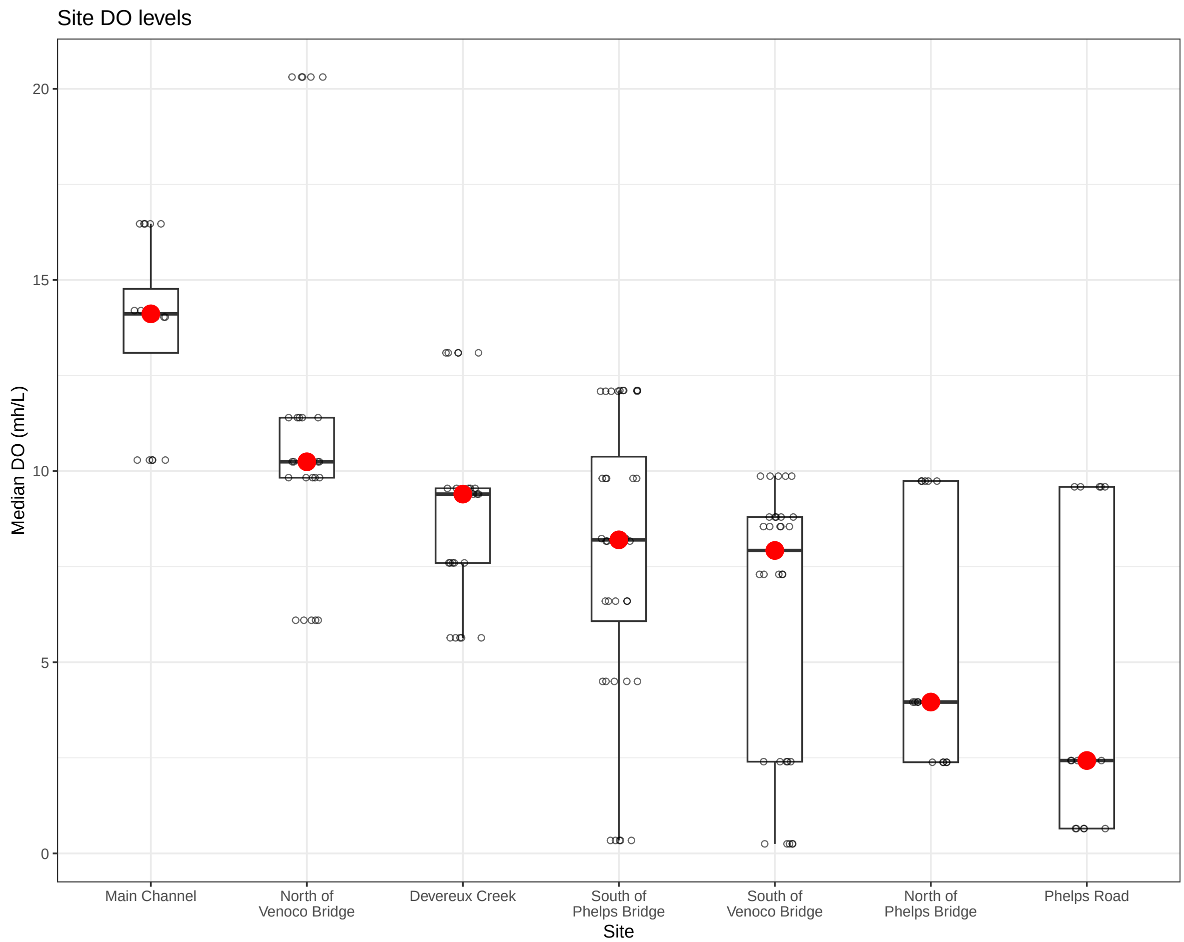
This figure helps address whether invertebrate abundance differs across dissolved oxygen levels by grouping continuous DO values into low, medium, and high categories. It shows that total invertebrate abundance may be higher under low-to-medium DO conditions and lower under high DO conditions in this dataset.

Other exploratory visualizations

Exploratory Figure 1: Variance of DO across sights

```
# create object aq_ins_clean_fig2 from aq_ins_clean
aq_ins_clean_fig2 <-  aq_ins_clean |>
  # convert taxon to title case
  mutate(taxon = str_replace_all(taxon, "_", " "),
         taxon = str_to_title(taxon)) |>
  # filter to include focus taxa
  filter(taxon %in% focus_taxa)

# base layer
DO <- ggplot(data = aq_ins_clean_fig2,
       # x-axis: site
       # y-axis: mean salinity
       mapping = aes(x = site_full,
                     y = med_do)) +
  # first layer: boxplot
  geom_boxplot(width = 0.35,
               outlier.shape = NA) +
  # second layer: salinity measurements at each survey
```

```
  geom_jitter(height = 0,
              width = 0.12,
              shape = 21,
              alpha = 0.6) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # second layer: points to represent means
  stat_summary(fun = median,
               color = "red",
               size = 1) +
  # relabelling x- and y-axes
  labs(x = "Site",
       y = "Median DO (mh/L)",
       title = "Site DO levels") +
  # different theme
  theme_bw()
# show plot
DO
```



Description of visual components

This figure shows how median DO varies across sampling sites. Each jittered open point represents one observation of median DO at a site, based on the focus invertebrate sampling data. The boxplots summarize the distribution of DO values at each site, where the box shows the IQR and the horizontal line inside the box shows the median. The large red point marks the median DO value for each site.

Main message / takeaway

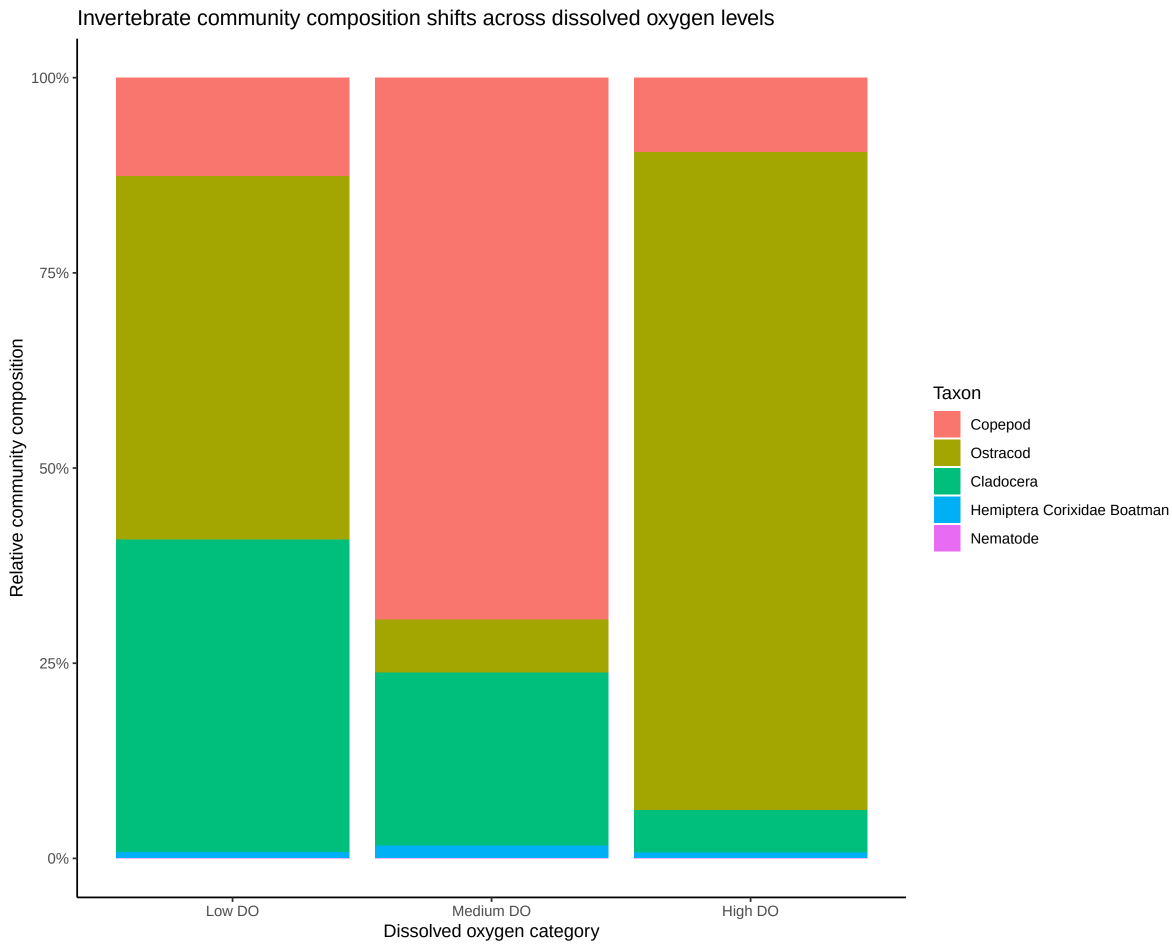
There are clear differences in dissolved oxygen among sites. Main Channel has the highest DO values overall, while North of Phelps Bridge and Phelps Road tend to have the lowest DO values, with the other sites showing intermediate conditions.

Connection to the main question

This figure helps contextualize the main question by showing that dissolved oxygen is not uniform across the study area, and varies substantially by site. These differences in DO between sites may help explain patterns in invertebrate abundance, since organisms at different sites are experiencing different oxygen conditions.

Exploratory Figure 2: Composition plot across DO bins

```
composition_plot <- DO_taxa |>
  mutate(
    do_category = case_when(
      med_do <= quantile(med_do, 0.25, na.rm = TRUE) ~ "Low DO",
      med_do <= quantile(med_do, 0.75, na.rm = TRUE) ~ "Medium DO",
      TRUE ~ "High DO"
    ),
    do_category = factor(do_category,
                         levels = c("Low DO",
                                    "Medium DO",
                                    "High DO"))
  ) |>
  group_by(do_category, taxon) |>
  summarize(
    total_abundance = sum(ave_lit, na.rm = TRUE),
    .groups = "drop"
  ) |>
  ggplot(aes(x = do_category,
             y = total_abundance,
             fill = taxon)) +
  geom_col(position = "fill") +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(
    x = "Dissolved oxygen category",
    y = "Relative community composition",
    fill = "Taxon",
    title = "Invertebrate community composition shifts across dissolved oxygen levels"
  ) +
  theme_classic()

# display plot
composition_plot
```



Description of visual components

This figure shows how relative invertebrate community composition changes across three dissolved oxygen categories: Low DO, Medium DO, and High DO. Each stacked bar represents the total abundance of the focus taxa within one DO category, scaled to 100% so that the bar shows relative composition rather than raw abundance. Each colored segment within a bar represents the proportion contributed by one focus taxon and the y-axis shows the percent of the community made up by each group.

Main message / takeaway

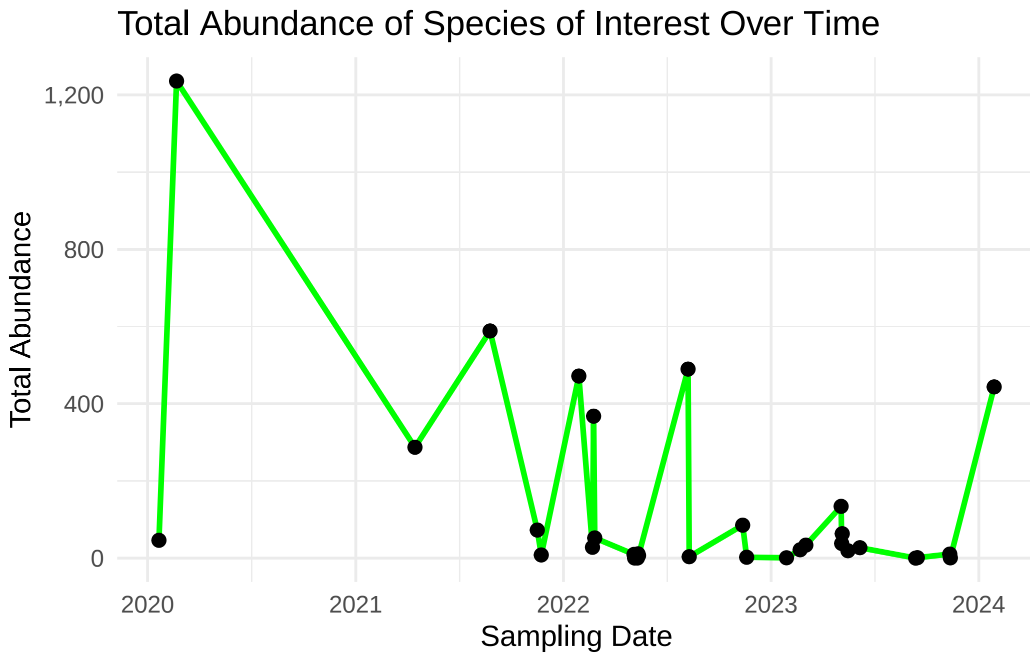
The figure suggests that invertebrate community composition shifts across dissolved oxygen levels. Low DO sites have a more mixed community with large contributions from Cladocera and Ostracods, Medium DO sites are dominated by Copepods, and High DO sites are dominated primarily by Ostracods, while Hemiptera Corixidae Boatman and Nematodes remain minor components across all DO levels.

Connection to the main question

This figure helps address the main question by showing that dissolved oxygen may influence not only total invertebrate abundance, but also which taxa make up the community. It provides important context by suggesting that changes in oxygen conditions are associated with shifts in community structure, meaning different invertebrate groups may respond to different DO levels.

Exploratory Figure 3: Time-series plot of total abundance (all species)

```
# create total abundance over time
total_abundance_time <- DO_taxa |>
  # keeps only the taxa of interest
  filter(taxon %in% focus_taxa) |>
  # makes sure date is treated as a date
  mutate(date = as.Date(date)) |>
  # groups by date only, so abundance is summed across sites and taxa
  group_by(date) |>
  # sums average abundance per liter across all taxa of interest
  summarise(
    total_abundance = sum(ave_lit, na.rm = TRUE),
    .groups = "drop"
  )

# plot total abundance over time
ggplot(total_abundance_time, aes(x = date, y = total_abundance)) +
  # adds a line connecting total abundance values over time
  geom_line(linewidth = 1,
            color = 'green') +
  # adds points for each sampling date
  geom_point(size = 2) +
  # formats y-axis labels with commas
  scale_y_continuous(labels = scales::comma) +
  # adds clear labels
  labs(
    title = "Total Abundance of Species of Interest Over Time",
    x = "Sampling Date",
    y = "Total Abundance"
  ) +
  # uses a clean theme
  theme_minimal()
```



Description of visual components

This figure shows total abundance of the focus invertebrate taxa over time. Each black point represents one sampling date, where abundance was summed across all sites and across all taxa of interest. The green line connects sampling dates in chronological order to show changes in total abundance over time. The x-axis shows sampling date, and the y-axis shows total abundance.

Main message / takeaway

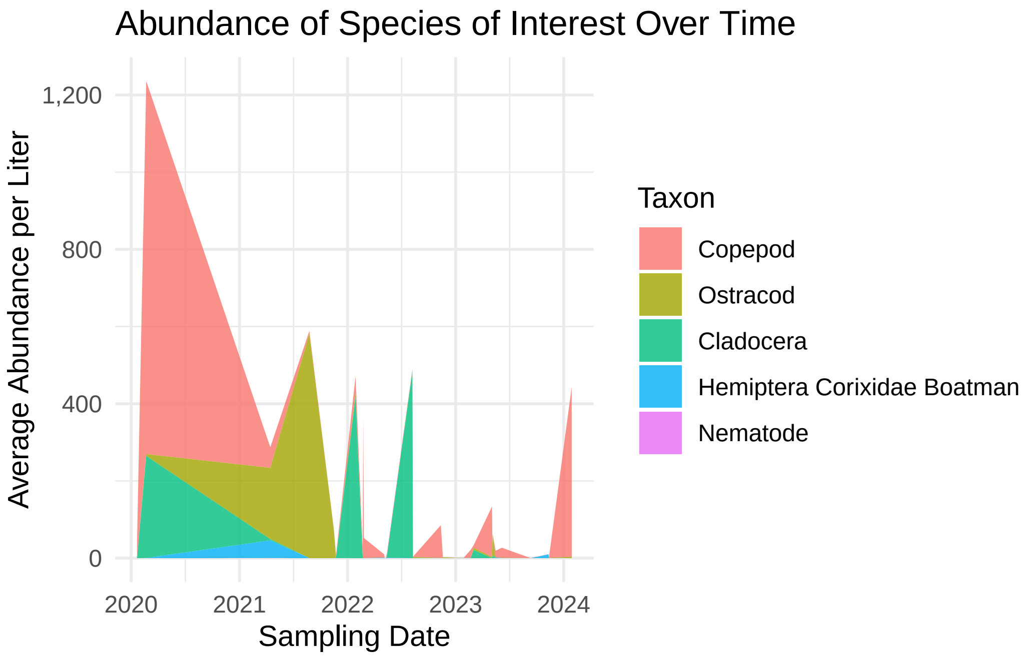
Total invertebrate abundance varies substantially over time, with one very large peak in 2020 and several smaller peaks in later years. There is no consistent increasing or decreasing trend across the full time period. Abundance appears highly variable among sampling dates.

Connection to the main question

This figure provides context for the dissolved oxygen and invertebrate abundance relationship by showing that abundance changes strongly across sampling dates. This is important because variation over time may influence observed abundance patterns in addition to dissolved oxygen levels.

Exploratory Figure 4: Time-series area plot

```r
# create abundance over time by taxon
taxon_abundance_time <- DO_taxa |>
  # keeps only the taxa of interest
  filter(taxon %in% focus_taxa) |>
  # makes sure date is treated as a date
  mutate(date = as.Date(date)) |>
  # groups by date and taxon so each species stays separate
  group_by(date, taxon) |>
  # sums across sites, but not across taxa
  summarise(
    ave_lit = sum(ave_lit, na.rm = TRUE),
    .groups = "drop"
  )

# create time series area plot by taxon
ggplot(taxon_abundance_time, aes(x = date, y = ave_lit, fill = taxon)) +
  # creates stacked area plot with a different color for each taxon
  geom_area(alpha = 0.8) +
  # formats y-axis labels with commas
  scale_y_continuous(labels = scales::comma) +
  # adds clear labels
  labs(
    title = "Abundance of Species of Interest Over Time",
    x = "Sampling Date",
    y = "Average Abundance per Liter",
    fill = "Taxon"
  ) +
  # uses a clean theme
  theme_minimal()
```



Description of visual components

This figure shows changes in abundance over time for our five focus invertebrate taxa: Ostracod, Copepod, Cladocera, Hemiptera Corixidae Boatman, and Nematode. Each colored area represents the summed average abundance per liter for one taxon on each sampling date, summed across sites but kept separate by taxon. The stacked height of the colored areas represents the total abundance of all focus taxa combined at that sampling date. The x-axis shows sampling date, and the y-axis shows average abundance per liter.

Main message / takeaway

Abundance varies strongly over time, and different taxa contribute to peaks at different points. Copepods appear to drive the largest peak in 2020 and another increase near 2024, while Ostracods and Cladocera contribute noticeably to some intermediate peaks.

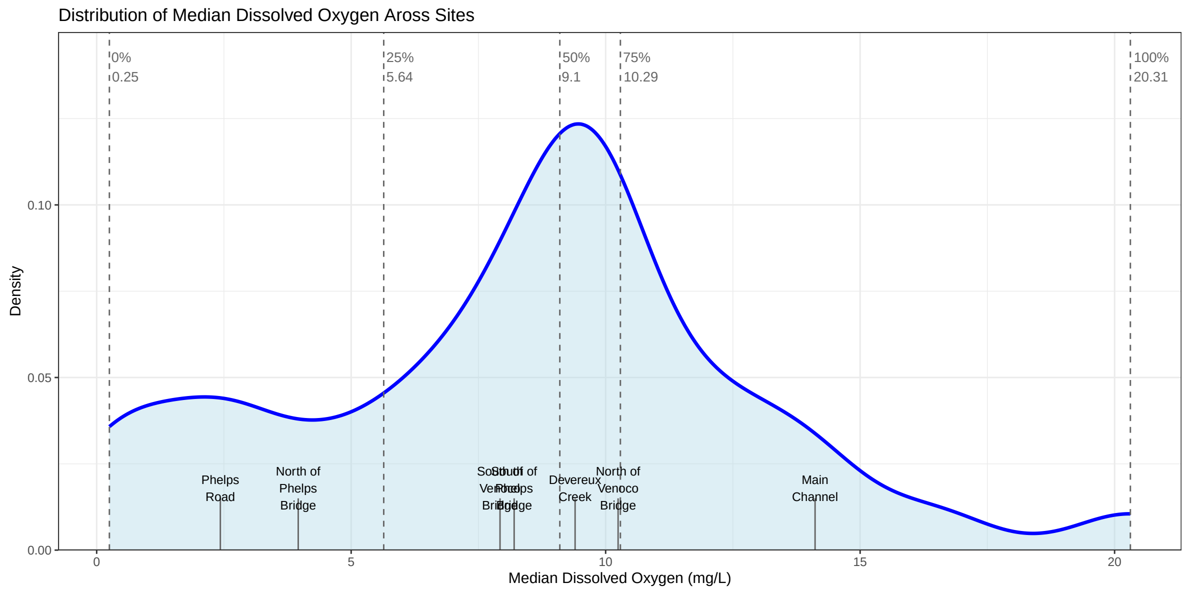
Connection to the main question

This figure adds temporal and taxon-specific context to the relationship between dissolved oxygen and invertebrate abundance. It shows that abundance patterns are not constant over time and that changes in total abundance may be driven by shifts in particular taxa, which is important when interpreting how the community responds to DO conditions.

Exploratory Figure 5: Distribution of Median Dissolved Oxygen Aross Sites

```
# create new object to store medians of median do levels
median_fig2 <- aq_ins_clean_fig2 |>
  group_by(site_full) |>
  summarize(median_do = median(med_do, na.rm = TRUE))

# assign quartiles for each site
aq_ins_clean_fig2$quartile <- cut(
  aq_ins_clean_fig2$med_do,
  breaks = quantile(aq_ins_clean_fig2$med_do, na.rm = TRUE, probs = seq(0, 1, 0.25)),
  labels = c("Q1", "Q2", "Q3", "Q4"),
  include.lowest = TRUE,
  right = FALSE)

# boundaries of quartiles
q <- quantile(aq_ins_clean_fig2$med_do, na.rm = TRUE, probs = seq(0, 1, 0.25))

view(q)

dens <- density(aq_ins_clean_fig2$med_do, na.rm = TRUE)

# one row per site
label_df <- aq_ins_clean_fig2 |>
  group_by(site_full) |>
  summarize(median_do = median(med_do, na.rm = TRUE))

# approximate density height for each site
label_df$y <- approx(
  dens$x,
  dens$y,
  xout = label_df$median_do
)$y

# distribution plot of DO levels across sites
ggplot(aq_ins_clean_fig2, aes(x = med_do)) +
  # density curve
  geom_density(fill = "lightblue",
               alpha = 0.4,
               color = "blue",
               linewidth = 1.2) +
  # vertical quartile lines
```

```
geom_vline(xintercept = q,
           linetype = "dashed",
           color = "gray40") +
# segments pointing to sites
 geom_segment(
  data = label_df,
  aes(
    x = median_do,
    xend = median_do,
    y = 0,
    yend = 0.015
  ),
  color = "gray40"
) +
# site labels
geom_text(
  data = label_df,
  aes(
    x = median_do,
    y = 0.018,
    label = str_wrap(site_full, width = 10)
  ),
  size = 3
) +
# quartile dashed line labels
geom_text(data = data.frame(x = q, label = names(q)),
          aes(x = x, y = 0.14, label = paste0(label, "\n", round(x, 2))),
          color = "gray40",
          size = 3.5,
          hjust = -0.1) + # shift text slightly to the right of the line
# remove space between bottom of plot and y-axis, set y-axis limits
scale_y_continuous(expand = c(0,0),
                   limits = c(0, 0.15)) +
# axes labels and title
labs(x = "Median Dissolved Oxygen (mg/L)",
     y = "Density",
     title = "Distribution of Median Dissolved Oxygen Aross Sites") +
# theme
theme_bw() +
# remove legend
theme(legend.position = "none")
```



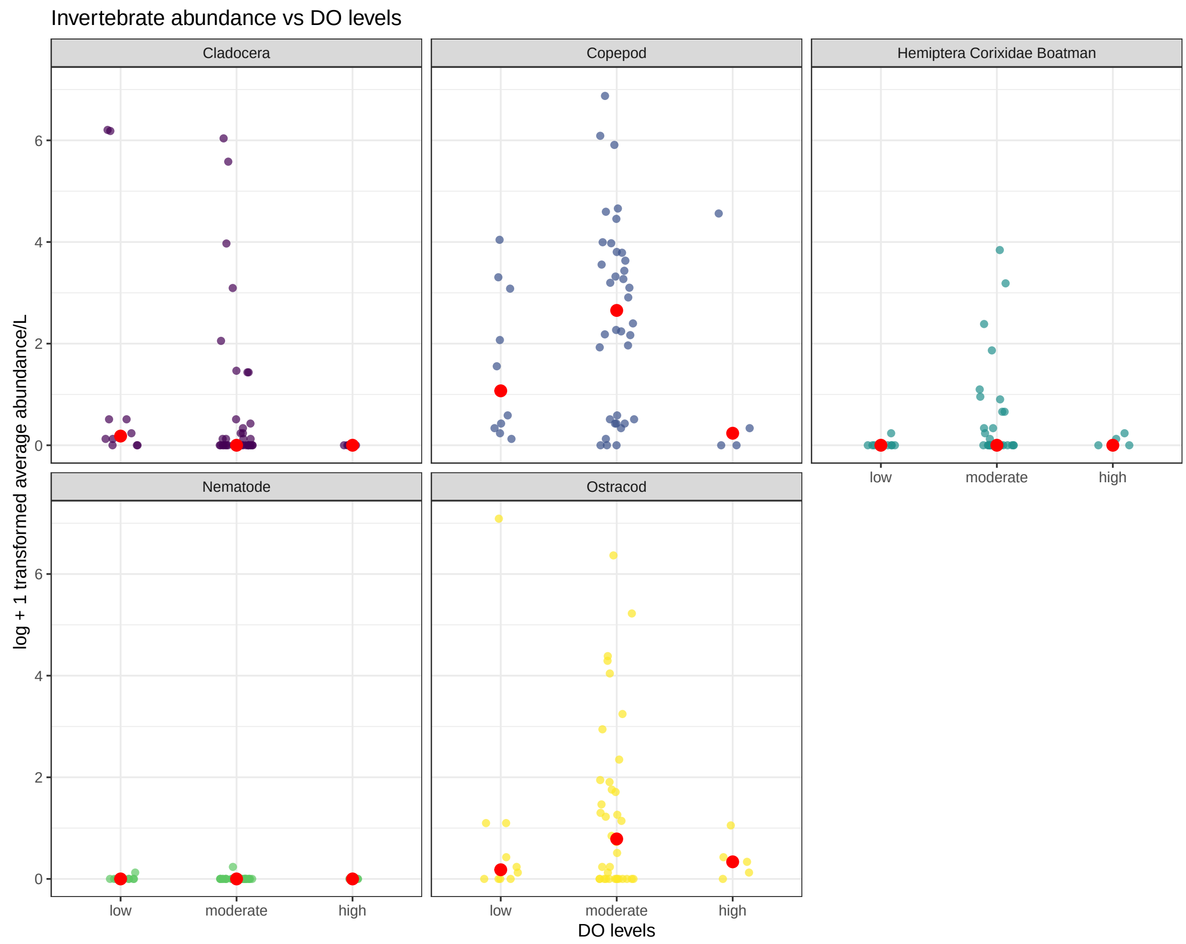
These bins were created by according to the median dissolved oxygen levels at each site. Low DO is the 25th percentile, moderate DO is the 75th percentile, and high DO is the 100th percentile. The labels show the median of the median DO measurments at each site.

- Low DO: Phelps Road and North of Phelps Bridge
- Moderate DO: South of Venoco Bridge, South of Phelps Bridge, Devereux Creek, North of Venoco Bridge
- High DO: Main Channel

Exploratory Figure 6: Invertebrate Abundance in DO Bins

```
aq_ins_clean_bins <- aq_ins_clean |>
  mutate(bins = case_when(site_full == "Phelps Road" ~ "low",
                          site_full == "North of Phelps Bridge" ~ "low",
                          site_full == "South of Venoco Bridge" ~ "moderate",
                          site_full == "South of Phelps Bridge" ~ "moderate",
                          site_full =="Devereux Creek" ~ "moderate",
                          site_full =="North of Venoco Bridge" ~ "moderate",
                          site_full == "Main Channel" ~ "high")) |>
  mutate(taxon = str_replace_all(taxon, "_", " "), taxon = str_to_title(taxon)) |>
  filter(taxon %in% focus_taxa) |>
  # create column that calculates log-transformed abundance
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  mutate(bins = factor(bins, levels = c("low", "moderate", "high")))
```

```r
# base layer: ggplot
ins_bins_plot <- ggplot(data = aq_ins_clean_bins,
                        mapping = aes(x = bins, # x-axis: DO bins
                                      y = log_ave_lit, # y-axis: log_ave_lit
                                      color = taxon)) + # color by taxon
  # first layer: points show individual abundance measurements
  geom_jitter(width = 0.15,
              height = 0,
              shape = 16,
              size = 2,
            alpha = 0.7) +
  # calculate median for each bin for each taxon
  stat_summary(fun = median,
               geom = "point",
               color = "red",
               size = 3) +
  # separate panels for each taxon
  facet_wrap(~ taxon) +
  # use non-default colors for points
  scale_color_viridis_d() +
  labs(x = "DO levels", # relabel x-axis
       y = "log + 1 transformed average abundance/L", # relabel y-axis
       # creates descriptive title
       title = "Invertebrate abundance vs DO levels") +
  # uses non-default theme
  theme_bw() +
  # remove redundant legend
  theme(legend.position = "none")

# shows plot
ins_bins_plot
```



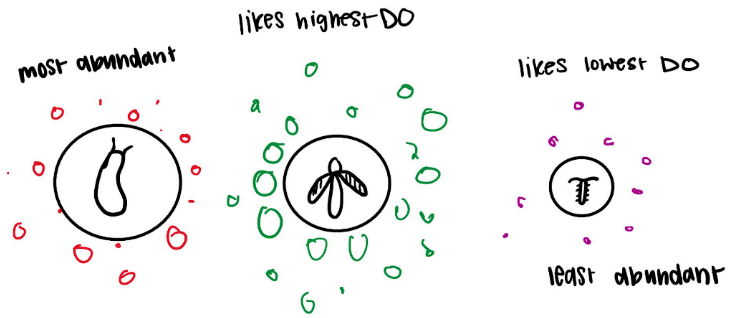
Plan for elective (if completing)

- should include some description/image of drafts
- should include plan for making progress in the next two weeks

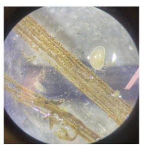
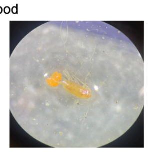
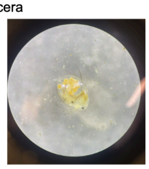
Elective idea: add microscopic images of each of the 5 main inverts

Water color bubbles/ blow colored bubbles with watercolor around each to demonstrate how much dissolved oxygen each likes

- More bubbles = likes more DO
- Less bubbles = likes less DO



Most common inverts

1. Ostracod
    a. 
2. Copepod
    a. 
3. Cladocera
    a. 
4. Corixidae (Water Boatmen)
    a. 
5. Nematode

Plan for next 2 weeks:

NEED TO ADD